



AI INNOVATION HACKATHON 2026

Explainable AI (XAI)

STANDARDIZED PROBLEM STATEMENTS

Women in AI, Research, Innovation & Entrepreneurship Club

Research • Innovation • Entrepreneurship

6 Challenges • Online • AI Innovation Hackathon 2026

Each challenge follows the same structured format for clarity, consistency and ease of review.

CHALLENGE PORTFOLIO

NO.	CHALLENGE	DIFFICULTY
1	Explain My Learning Risk	Intermediate
2	Explain My Loan Decision	Intermediate
3	Explain the Health Risk Prediction	Intermediate
4	Explain Which Public Complaint Needs Priority	Intermediate
5	Explain Why a Message Looks Suspicious	Intermediate
6	Explain an AI Decision Affecting People	Intermediate to Advanced



CHALLENGE 01

Explain My Learning Risk

DOMAIN

Education / Student Support

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Educational institutions may use data to identify students at risk of poor performance, but an unexplained prediction can be difficult for teachers and students to trust or act upon.

EXPECTED SOLUTION

Build a student-performance prediction prototype. It should accept student information, predict a performance/risk category, identify the main factors influencing that result, and present those factors in simple educational language.

SUGGESTED DATASETS / RESOURCES

UCI Student Performance; UCI Student Academics Performance dataset.

AI / ML COMPONENT

Classification or regression using logistic regression, decision tree, random forest or gradient boosting; SHAP, LIME, permutation importance or an interpretable model.

WHAT MAKES THE CHALLENGE INTERESTING?

The key question is whether AI can predict student performance and explain why, while making explanations useful for possible intervention.

SUGGESTED ON-SCREEN DEMONSTRATION

Enter student information predicted category/risk local explanation global feature importance optional what-if view.

TECHNICAL REQUIREMENTS

Python; preprocessing; ML model; evaluation metrics; XAI; explanation visualization; simple interface/notebook; README.

SUGGESTED MINIMUM DELIVERABLE

- Working model and reproducible code/notebook.
- At least one local XAI explanation and one global explanation.
- Simple visualization, dashboard, or user-facing explanation.
- README covering data, model, XAI, evaluation, limitations and ethics.

OPTIONAL ADVANCED FEATURES

- Counterfactual explanations.
- Comparison of two XAI methods.
- Comparison of interpretable and black-box models.
- Fairness/subgroup analysis where appropriate.
- Interactive dashboard or natural-language explanation.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Do not present predictions as certain judgments about individual students. Avoid sensitive or unnecessary personal data, document dataset bias/limitations, and communicate that explanations indicate model behavior rather than causal facts.

INNOVATION ANGLE

Connects prediction with understandable educational explanations rather than treating a risk score as the final output.



SUCCESS / EVALUATION METRICS

Predictive performance plus explanation consistency, user comprehensibility, subgroup/error analysis where appropriate.



CHALLENGE 02

Explain My Loan Decision

DOMAIN

Finance / Responsible Lending

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Automated credit systems can classify applicants, but applicants and financial staff may need understandable reasons for a decision.

EXPECTED SOLUTION

Build a credit-risk prototype. For an individual application, show the model output, the most influential factors, whether each factor pushed the output toward or away from the result, and a human-readable explanation.

SUGGESTED DATASETS / RESOURCES

UCI Statlog German Credit Data.

AI / ML COMPONENT

Logistic regression, decision tree, random forest or gradient boosting; SHAP/LIME or feature importance for local and global explanations.

WHAT MAKES THE CHALLENGE INTERESTING?

Teams address both prediction and explanation; an optional counterfactual-style view can show which inputs would need to change for the model output to change without claiming causation.

SUGGESTED ON-SCREEN DEMONSTRATION

Enter application data model output factors pushing toward/away from result human-readable explanation optional counterfactual.

TECHNICAL REQUIREMENTS

Python; tabular preprocessing; classification; accuracy/precision/recall/F1 or ROC-AUC where appropriate; XAI; dashboard; basic fairness discussion.

SUGGESTED MINIMUM DELIVERABLE

- Working model and reproducible code/notebook.
- At least one local XAI explanation and one global explanation.
- Simple visualization, dashboard, or user-facing explanation.
- README covering data, model, XAI, evaluation, limitations and ethics.

OPTIONAL ADVANCED FEATURES

- Counterfactual explanations.
- Comparison of two XAI methods.
- Comparison of interpretable and black-box models.
- Fairness/subgroup analysis where appropriate.
- Interactive dashboard or natural-language explanation.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

This is not a real lending decision system. Avoid sensitive personal data, document fairness limitations, and do not interpret feature importance as proof of causation.

INNOVATION ANGLE

Combines local explanation, global behavior and optional counterfactual reasoning for a high-impact decision context.



SUCCESS / EVALUATION METRICS

Classification performance, fairness metrics where appropriate, explanation consistency, counterfactual validity if used.



CHALLENGE 03

Explain the Health Risk Prediction

DOMAIN

Healthcare / Clinical Decision Support

DIFFICULTY

Intermediate

PROBLEM STATEMENT

AI can identify patterns associated with health risks, but unexplained predictions are difficult to interpret. The prototype must be clearly presented as research/demo software, not medical diagnosis.

EXPECTED SOLUTION

Build a public-dataset health-risk classifier. For each prediction, show influential factors and an overall view of important features across the model.

SUGGESTED DATASETS / RESOURCES

UCI Heart Disease dataset.

AI / ML COMPONENT

Logistic regression, decision tree, random forest or gradient boosting; SHAP, LIME, permutation importance or another suitable XAI method.

WHAT MAKES THE CHALLENGE INTERESTING?

The high-stakes setting makes explanation central. Teams should show what influenced the output and what the model does not know.

SUGGESTED ON-SCREEN DEMONSTRATION

Enter sample risk prediction influential factors global importance optional small input change comparison.

TECHNICAL REQUIREMENTS

Python; preprocessing; suitable medical classification metrics; XAI; visualization/dashboard; responsible-AI statement; no private patient data.

SUGGESTED MINIMUM DELIVERABLE

- Working model and reproducible code/notebook.
- At least one local XAI explanation and one global explanation.
- Simple visualization, dashboard, or user-facing explanation.
- README covering data, model, XAI, evaluation, limitations and ethics.

OPTIONAL ADVANCED FEATURES

- Counterfactual explanations.
- Comparison of two XAI methods.
- Comparison of interpretable and black-box models.
- Fairness/subgroup analysis where appropriate.
- Interactive dashboard or natural-language explanation.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Research/demo only, not medical diagnosis. Do not use private patient data. Document dataset limitations, possible bias and uncertainty, and avoid causal claims.

INNOVATION ANGLE

Makes explanation and limitations a first-class output of a health-risk model.

SUCCESS / EVALUATION METRICS



Classification metrics, calibration if used, explanation stability, subgroup/error analysis where possible.



CHALLENGE 04

Explain Which Public Complaint Needs Priority

DOMAIN

Governance / Public Services

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Public organizations may receive many complaints. AI could classify, prioritize or route them, but opaque prioritization can reduce trust.

EXPECTED SOLUTION

Create a system accepting complaint text or structured data, predicting category/priority and recommending a department/queue. Explain which words or features influenced the result.

SUGGESTED DATASETS / RESOURCES

Use an open complaint/request dataset from a public-data portal, or create a small synthetic/curated dataset with documented labels.

AI / ML COMPONENT

TF-IDF with logistic regression/linear SVM, a tree-based model, or a lightweight text model; LIME/SHAP or model-specific feature weights.

WHAT MAKES THE CHALLENGE INTERESTING?

Combines NLP, XAI and public-service decision making; teams can show why a complaint was marked urgent and why it was routed to a department.

SUGGESTED ON-SCREEN DEMONSTRATION

Enter complaint category/priority recommended department highlighted words/features global feature view.

TECHNICAL REQUIREMENTS

Python; NLP preprocessing; text/structured classification; evaluation metrics; local explanation; global feature analysis; dashboard/API; privacy and bias considerations.

SUGGESTED MINIMUM DELIVERABLE

- Working model and reproducible code/notebook.
- At least one local XAI explanation and one global explanation.
- Simple visualization, dashboard, or user-facing explanation.
- README covering data, model, XAI, evaluation, limitations and ethics.

OPTIONAL ADVANCED FEATURES

- Counterfactual explanations.
- Comparison of two XAI methods.
- Comparison of interpretable and black-box models.
- Fairness/subgroup analysis where appropriate.
- Interactive dashboard or natural-language explanation.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Avoid exposing personal information in demonstrations. Document label quality, bias, false-priority risks and that the system supports rather than replaces accountable public-service decisions.

INNOVATION ANGLE

Combines text classification, prioritization and explanation in a public-service workflow.



SUCCESS / EVALUATION METRICS

Classification/prioritization metrics, routing accuracy, false-priority rate, explanation consistency and subgroup/error analysis.



CHALLENGE 05

Explain Why a Message Looks Suspicious

DOMAIN

Cybersecurity / Trustworthy Detection

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Spam and suspicious-message systems can block legitimate messages or miss harmful ones. Users and analysts need to know why a message was classified as suspicious.

EXPECTED SOLUTION

Build a spam/non-spam or suspicious/non-suspicious classifier. For every prediction, highlight words/features that influenced the result and provide a simple explanation.

SUGGESTED DATASETS / RESOURCES

UCI SMS Spam Collection; UCI Spambase.

AI / ML COMPONENT

For text: TF-IDF plus logistic regression/linear SVM, Naive Bayes, or a lightweight transformer; LIME, SHAP or model-specific feature importance.

WHAT MAKES THE CHALLENGE INTERESTING?

The explanation works at message level: users see why a message was flagged rather than only receiving a label. Teams can analyze false positives and model mistakes.

SUGGESTED ON-SCREEN DEMONSTRATION

Paste message suspicious/non-suspicious prediction highlighted influential words/features optional confidence explanation.

TECHNICAL REQUIREMENTS

Python; NLP; train/test split; precision, recall and F1; local explanations; highlighted text/visualization; optional confidence score; false-positive analysis.

SUGGESTED MINIMUM DELIVERABLE

- Working model and reproducible code/notebook.
- At least one local XAI explanation and one global explanation.
- Simple visualization, dashboard, or user-facing explanation.
- README covering data, model, XAI, evaluation, limitations and ethics.

OPTIONAL ADVANCED FEATURES

- Counterfactual explanations.
- Comparison of two XAI methods.
- Comparison of interpretable and black-box models.
- Fairness/subgroup analysis where appropriate.
- Interactive dashboard or natural-language explanation.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Use safe/public datasets and do not include real private messages or credentials. Clearly distinguish classification from certainty that a message is malicious.

INNOVATION ANGLE

Turns a binary security classification into an interpretable analyst/user-facing decision aid.



SUCCESS / EVALUATION METRICS

Precision, recall, F1, false-positive rate, explanation consistency and robustness to small text changes.



CHALLENGE 06

Explain an AI Decision Affecting People

DOMAIN

Law / Social Systems / Fairness

DIFFICULTY

Intermediate to Advanced

PROBLEM STATEMENT

AI may influence access to education, employment, public services or opportunities. People may need to understand which factors contributed to an automated decision.

EXPECTED SOLUTION

Create a decision-support model for a defined public-interest scenario such as admission screening, job shortlisting, or service access using public or synthetic data. Provide local, global and optionally counterfactual explanations.

SUGGESTED DATASETS / RESOURCES

Use a public tabular dataset relevant to the selected scenario, or construct a clearly documented synthetic dataset. UCI datasets may also be adapted for educational demonstrations.

AI / ML COMPONENT

Compare a transparent model such as logistic regression/decision tree with a more complex model such as random forest/gradient boosting; apply SHAP/LIME/permutation importance and optionally counterfactual explanations.

WHAT MAKES THE CHALLENGE INTERESTING?

The challenge asks whether an explanation is understandable and whether it changes appropriately when inputs change. Teams can discuss transparency, fairness and accountability without treating feature importance as causation.

SUGGESTED ON-SCREEN DEMONSTRATION

Select scenario model decision local explanation global explanation optional counterfactual fairness/subgroup view.

TECHNICAL REQUIREMENTS

Python; classification/regression; model comparison; local/global XAI; optional counterfactuals; fairness/subgroup analysis; dashboard/report; responsible-AI documentation.

SUGGESTED MINIMUM DELIVERABLE

- Working model and reproducible code/notebook.
- At least one local XAI explanation and one global explanation.
- Simple visualization, dashboard, or user-facing explanation.
- README covering data, model, XAI, evaluation, limitations and ethics.

OPTIONAL ADVANCED FEATURES

- Counterfactual explanations.
- Comparison of two XAI methods.
- Comparison of interpretable and black-box models.
- Fairness/subgroup analysis where appropriate.
- Interactive dashboard or natural-language explanation.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Use public or synthetic data for sensitive scenarios. Do not treat feature importance as causation. Document fairness, privacy, limitations and the difference between model behavior and real-world justification.

INNOVATION ANGLE

Explicitly compares interpretability, performance and accountability in a public-interest decision context.



SUCCESS / EVALUATION METRICS

Model performance, explanation quality/consistency, fairness metrics where applicable, counterfactual validity if used, and usability.